

Sample Project Flow

Worked Example: Diabetic Retinopathy Grading (APTOS 2019)

How to use this: One end-to-end example so teams see exactly what each stage looks like. Numbers shown are placeholders - teams fill in their own real results.

Stage 1 - Literature survey

Search: “diabetic retinopathy grading deep learning APTOS”. Read 10-20 recent papers. For each, note: which backbone, how they handled the severe class imbalance, what preprocessing, and which metric (APTOS uses quadratic weighted kappa, QWK). Common pattern: transfer-learning CNN (ResNet/EfficientNet) fine-tuned on APTOS, accuracy + QWK reported, minority grades (Severe, Proliferative) hardest.

Stage 2 - Select base paper + reproducibility check

Pick ONE transfer-learning DR-grading paper with public code and results on APTOS 2019. Fill the checklist:

Checklist item	Team's answer (example)
Public code?	Yes - GitHub repo linked in paper (Preferred)
Dataset public & size?	APTOS 2019, 3,662 images, ~9 GB - downloadable from Kaggle
Fits free GPU?	Yes at 224×224, batch 16–32 on Colab/Kaggle
Last commit / usable?	Repo runs; minor dependency fixes only
Metric defined?	QWK (primary) + accuracy + per-class recall

Stage 3 - Reproduce the baseline

Re-run the paper's method on APTOS with the **same protocol** (same split logic, same preprocessing, same backbone) and report QWK + accuracy + a per-class recall table. Goal: land *near* the paper's number. If the paper reports QWK ≈ 0.90 and you get ≈ 0.86 , record the gap - it becomes analysis in Stage 6 (different seed? preprocessing? split?).

Stage 4 - Propose a hypothesis (BEFORE coding)

Example hypothesis: “APTOS is severely imbalanced (few Severe/Proliferative cases). Replacing cross-entropy with focal loss will raise recall on the minority grades and improve QWK, because focal loss down-weights easy majority examples and focuses learning on hard minority ones. Predicted: minority-class recall +8–15%, QWK +0.02–0.05; overall accuracy may dip slightly.”

Note the shape: one variable (loss function), a direction, a rough magnitude, and a mechanism. Alternative single-variable hypotheses: CLAHE contrast preprocessing; backbone swap ResNet50 → EfficientNet-B0; class-weighted sampling.

Stage 5 - Implement & run ablations

Change ONE thing at a time and fill a table like this (numbers are placeholders):

Configuration	QWK	Minority recall	Notes
Baseline (cross-entropy)	0.____	____%	Reproduced baseline
+ Focal loss (hypothesis)	0.____	____%	One variable changed
+ CLAHE preprocessing	0.____	____%	Second experiment (optional)
EfficientNet-B0 backbone	0.____	____%	Backbone ablation (optional)

For each row: predict first, run, then compare reality to your prediction. The gap between expected and actual is where the learning (and the marks) live.

Stage 6 - Failure analysis & write-up

- **Confusion matrix:** which grades get confused? (Often adjacent grades, e.g., Mild vs. Moderate.)
- **Per-class recall:** did focal loss actually help the minority grades, as predicted?
- **Failure images:** show 5–10 misclassified fundus images; hypothesise why (blur, poor illumination, subtle lesions).
- **Reproduction gap:** explain any difference from the paper's number.
- **Limitations:** single dataset, no external validation, NO clinical claim.

Write it in the **IEEE conference template** - sections: Abstract, Introduction, Related Work (from Stage 1), Method (baseline + your change), Experiments (the ablation table), Results & Discussion (including the failure analysis and reproduction gap), Limitations, Conclusion, References.

What “good” looks like: A team that predicted focal loss would help minority grades, tested it, found it helped recall but slightly hurt overall accuracy, explained WHY with the confusion matrix, and honestly discussed the reproduction gap - has done real research, entirely on free Colab, with a pretrained backbone.